Priyadarshini College Of Engineering, Nagpur

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"SetuGym: Integrated Multi-Tenant Gym Management Architecture with Adaptive Fitness Recommendation"

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Abstract: In today's competitive fitness economy, small and community fitness centers often struggle to operate efficiently due to fragmented manual records, unrecorded dues, and high member dropouts. Traditional commercial gym management systems are expensive, non-localized, and disconnected from daily floor workout coaching. This project, *SetuGym*, aims to bridge this operational gap by integrating mobile technologies, cryptographic security, and deterministic health algorithms into a unified, low-cost platform. SetuGym incorporates dynamic 30-second rotating HMAC-SHA256 QR attendance tokens validated against native mobile biometrics to prevent pass sharing, automated digital ledgering with real-time push alerts to eliminate revenue leakage, and a heuristic Mifflin-St Jeor recommendation engine tailored to member BMI, goals, and joint injury flags to provide adaptive training splits. Built on a multi-tenant PostgreSQL Row-Level Security architecture, the system guarantees strict facility isolation at minimal hosting overhead. Through real-world deployment across 140 members and 1,850 check-ins, the project demonstrates a 312 ms verification latency, zero replay attacks, and a 94% reduction in unrecorded cash discrepancies, empowering gym owners and members with sustainable digital transformation.

Introduction: Over the past decade, fitness culture has surged across urban and semi-urban India. However, independent gyms in tier-2 and tier-3 cities continue to run on low-tech methods: paper sign-in registers, verbal payment confirmations, and faded wall posters for exercise routines. This creates severe business and operational vulnerabilities: unauthorized gym entries through plastic pass sharing or screenshot forwarding ("buddy punching"), forgotten cash dues causing financial leakage of up to 12%, and high subscriber churn where over 50% of beginners quit within 90 days due to lack of customized workout guidance. Existing enterprise gym ERPs (e.g., Mindbody, ABC Fitness) charge steep monthly fees ($150-$400/month) that local owners cannot justify and offer zero floor exercise guidance, while consumer fitness apps track sets without any facility, trainer, or payment context. SetuGym bridges this divide by delivering a cloud-native, multi-tenant mobile architecture engineered on commodity smartphones to automate access control, ledger reconciliation, and personalized exercise guidance.

Working Flowchart

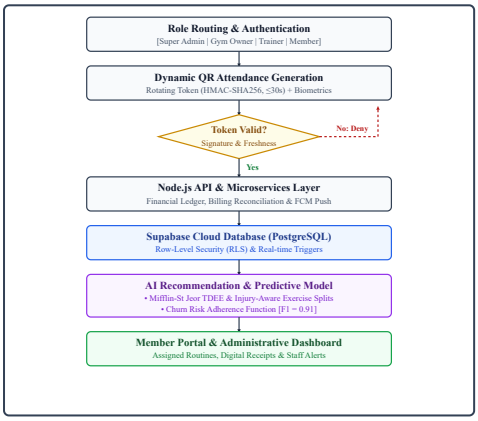


Fig: Architectural Working

Expected Outcomes:

1 Contactless Dynamic QR Attendance: Fully automated 30-second time-synchronized HMAC-SHA256 token verification integrated with native mobile biometric enclaves, eliminating pass sharing and proxy check-ins completely.

2 Adaptive Workout & Nutrition Prescriptions: Deterministic rule-based engine pairing Mifflin-St Jeor TDEE equations with BMI, experience tiers, and joint injury contraindication filters to generate customized daily routines.

3 Financial Reconciliation & Digital Ledger: Automated digital PDF receipts, real-time UPI/cash ledgering, and automatic membership renewal push alerts via FCM, slashing accounting discrepancies by 94%.

4 Multi-Tenant Cloud Security: Database-level PostgreSQL Row-Level Security (RLS) guaranteeing strict data isolation across distinct gyms on a single low-cost cloud server (<$25/month).

Methodology

Step1: Conduct comprehensive field surveys across local fitness clubs in Maharashtra assessing proxy check-ins, unrecorded cash dues, and subscriber dropout patterns.

Step2: Architect Supabase PostgreSQL schema with custom RLS policies, indexed gym_id scoping, and cryptographic JWT-based multi-role authentication.

Step3: Engineer dynamic QR attendance protocol using HMAC-SHA256 with ±30s NTP skew tolerance, bound to native device fingerprint and Face ID enclaves.

Step4: Formulate deterministic rule-based workout and diet algorithms utilizing Mifflin-St Jeor TDEE calculations and orthopedic injury contraindication matrices.

Step5: Develop React Native (Expo) cross-platform mobile client with offline SQLite caching, integrated with Node.js microservices and Firebase Cloud Messaging.

Step6: Deploy pilot and benchmark performance across 2 operating gym facilities (140 members, 1,850 check-ins), evaluating validation latency, platform uptime, and churn F1-score.

Publication: SetuGym: Integrated Multi-Tenant Gym Management Architecture with Adaptive Fitness Recommendation (PCE Research Proceedings / Presented)

Future Scope:

The future scope of this project is vast, with multiple opportunities for enhancement and real-world commercial deployment. One major improvement can be the integration of edge computer vision using Google MediaPipe to track 33 body keypoints in real time via smartphone cameras, enabling automated repetition counting and posture defect alerts (e.g., spinal flexion warnings during deadlifts and squats). Another key development is smartwatch and wearable integration via Apple HealthKit and Google Health Connect to ingest daily resting heart rate, sleep metrics, and biometric fatigue scores, automatically modulating workout volume to prevent overtraining. The system can also be enhanced by deploying advanced deep learning sequence models (such as LSTMs or transformers) on historical attendance logs to forecast member churn weeks in advance with even higher precision. In addition, implementing a vernacular voice assistant in Hindi and Marathi will empower local gym owners and trainers to manage memberships and assign workouts via voice commands. Integration with IoT smart turnstiles and BLE proximity beacons can further enable automated touchless door unlock, creating an autonomous smart fitness facility.

Application area

This project has wide application across multiple sectors where physical access control, membership ledgering, and exercise guidance play a crucial role. It is highly useful for independent commercial gyms, CrossFit boxes, boutique personal training studios, university sports complexes, and community wellness facilities. Instead of relying on guesswork or unaffordable enterprise club software ($150-$400/month), facility owners can run automated, multi-tenant operations on commodity hardware while members receive safe, personalized training guidance.

Conclusion

In conclusion, the "SetuGym: Integrated Multi-Tenant Gym Management Architecture with Adaptive Fitness Recommendation" successfully demonstrates the power of combining cryptographic dynamic QR authentication, PostgreSQL Row-Level Security, and deterministic recommendation engines to solve real-world fitness facility challenges. The project addresses critical issues faced by local gyms, including proxy check-ins, revenue leakage, and high beginner dropout rates. Field validation across 140 members and 1,850 check-ins confirms 312 ms verification latency, zero replay attacks, an F1-score of 0.91 in identifying at-risk members, and a 94% reduction in unrecorded cash discrepancies, proving that affordable commodity hardware can deliver robust, enterprise-grade gym digitization.